



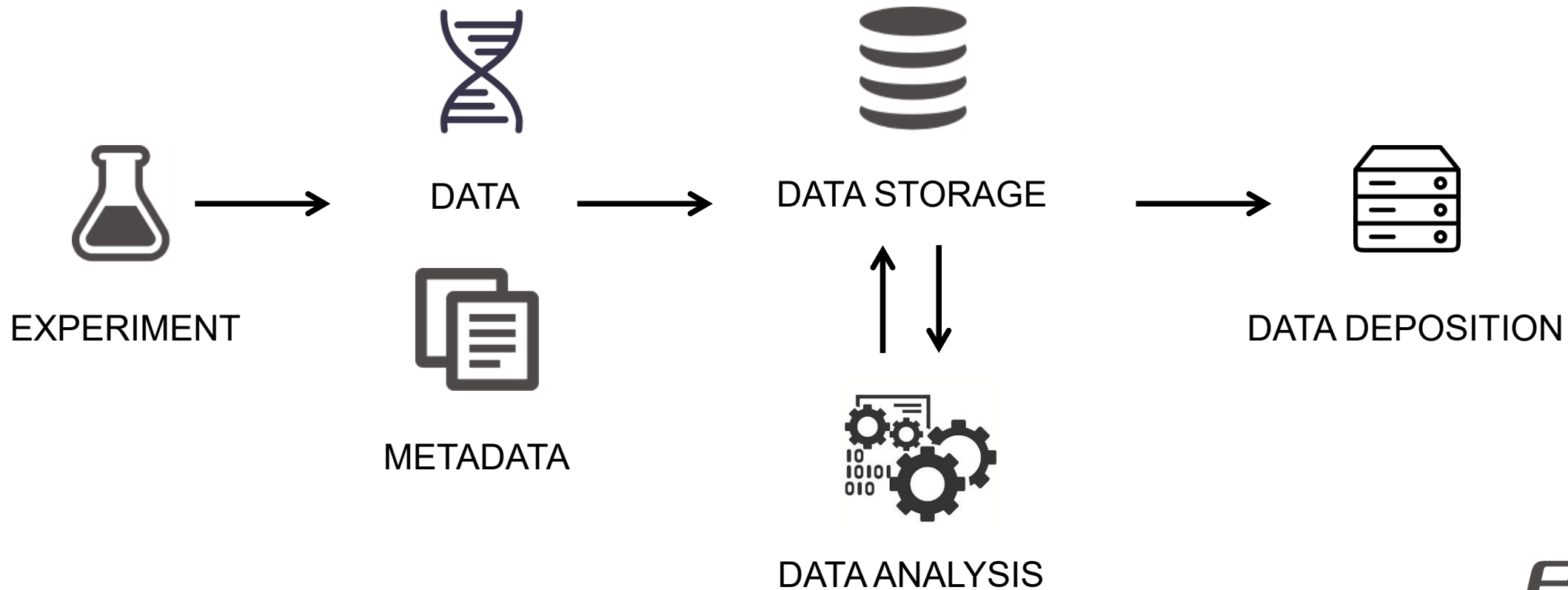
National storage infrastructures

NeLS/StoreBioInfo, NIRD, TSD

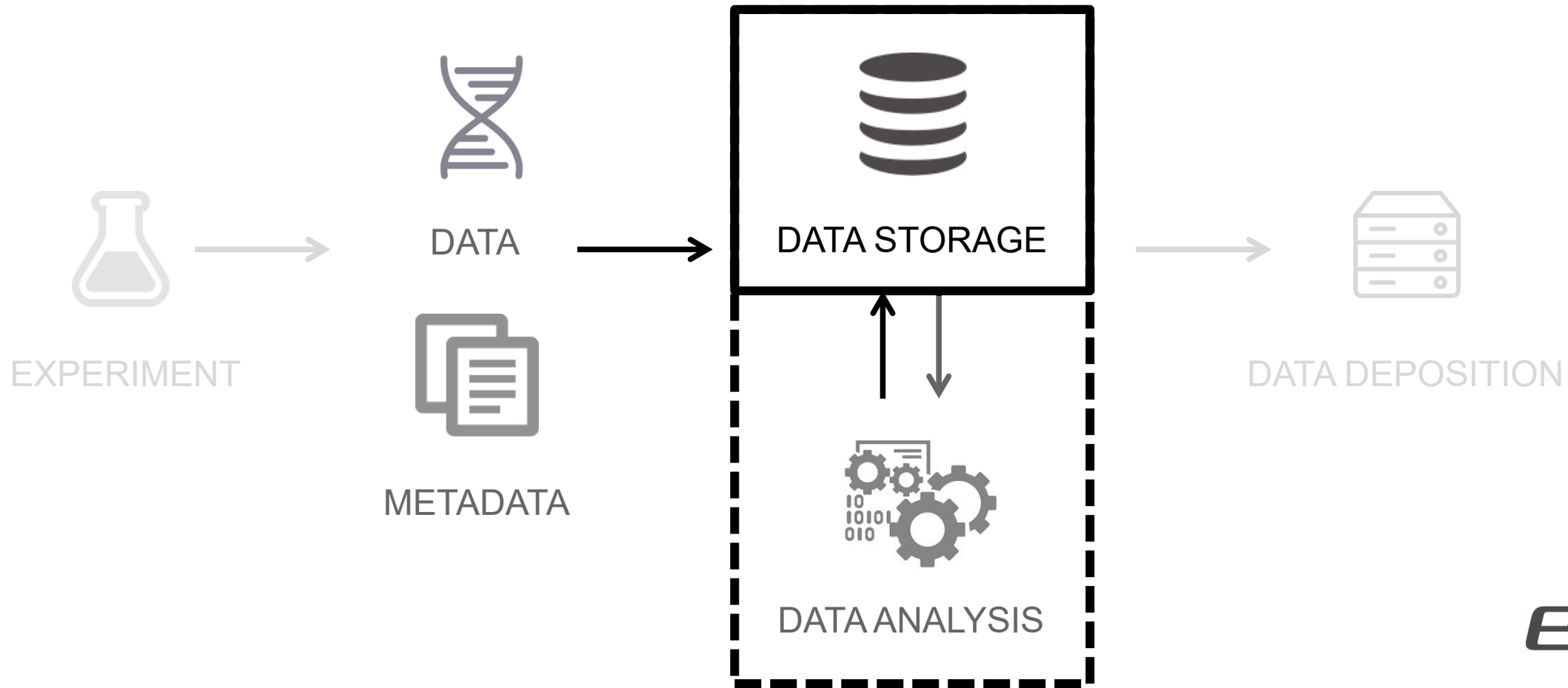


Erik Hjerde
Arturo Vera-Ponce de Leon
ELIXIR Norway

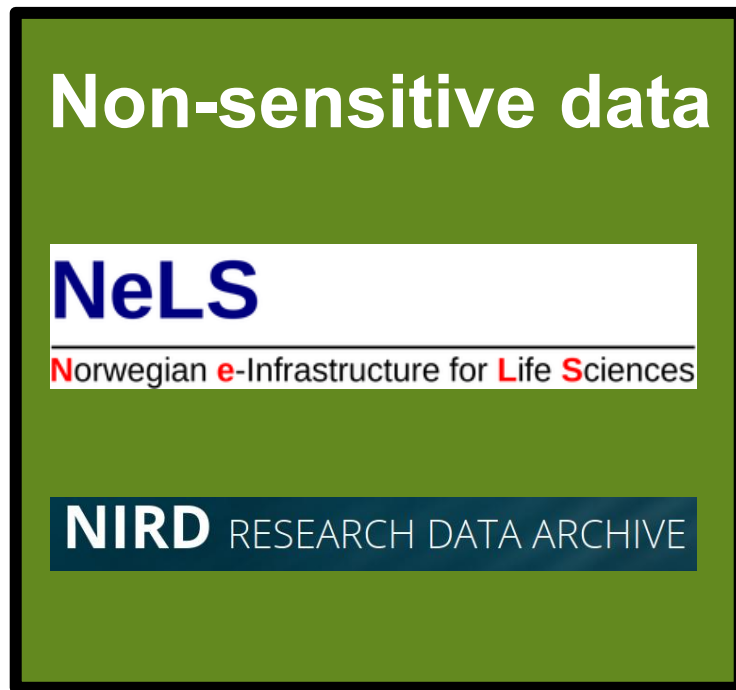
Data storage – from the researcher perspective



Data storage – from the researcher perspective



National storage infrastructures. Where to put my data???



DATA ANALYSIS



Data storage – General considerations

The ELIXIR Research Data Management kit

Data life cycle	+
Your role	+
Your domain	+
Your problem	-

Compliance monitoring

Data analysis

Data management plan

Data organisation

Data protection

Data publication

Data quality

Data storage

Data transfer

Identifiers

Licensing

Documentation and metadata

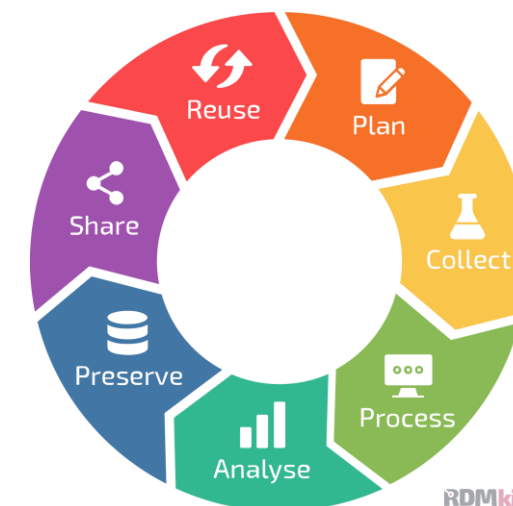
Sensitive data

All tools and resources

- What features do you need in a storage solution when collecting data?
- How do you estimate computational resources for data processing and analysis?
- Where should you store the data after the end of the project?
- Relevant tools and resources

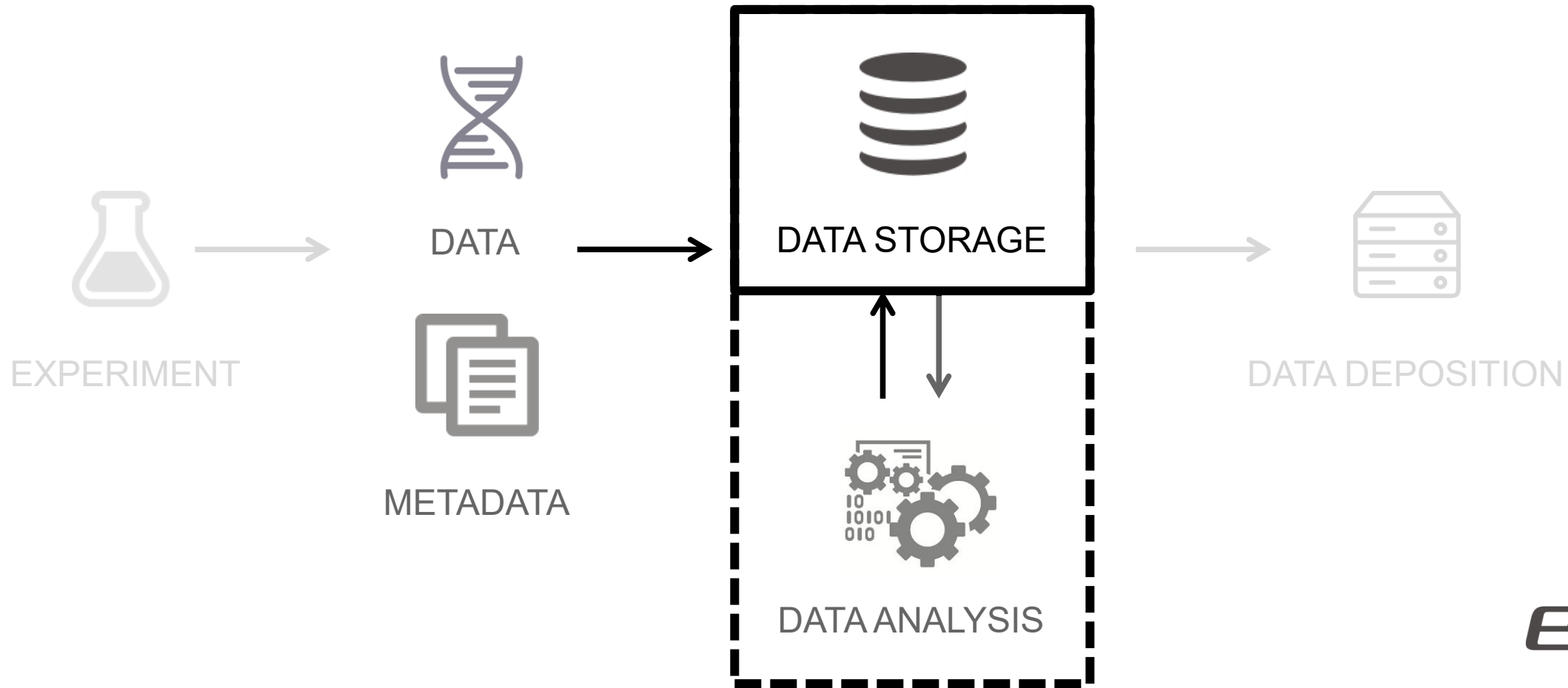
Relevant tools and resources

Tool or resource ⓘ	Description	Tags	Registry
CERNBox	CERNBox cloud data storage, sharing and synchronization	storage	
CS3	Cloud Storage Services for Synchronization and Sharing (CS3)	storage	
DATAVERSE	Open source research data repository software.	storage researcher data manager IT support	



Link to RDMkit storage: <https://rdmkit.elixir-europe.org/>

Data storage – from the researcher perspective



Elixir Norway



HOMEABOUTUSHELPDESKSERVICESTRAININGNEWSEVENTSCONTACT USCOVID-19 RESOURCES

Enter Search...

ELIXIR Norway

ELIXIR Norway is a national node of [ELIXIR](#), the pan-European infrastructure for biological information, supporting life science research and its translation to medicine, environment, the bioindustries and society.

ELIXIR Norway is funded by the Research Council of Norway, and builds upon the Norwegian Bioinformatics Platform, which has provided support and tools for Norwegian life science research for nearly two decades.



ELIXIR Norway is proud to be a part of ELIXIR-CONVERGE, to help standardise life science data management across Europe

News

[Life Science RDM Group event: The Data Steward job role in the Norwegian public sector](#)
Apr 24, 2024

[First datasets from Norway are now in the Federated EGA Network.](#)
Mar 4, 2024

[Sweden adds a National Pathogens Portal to the network](#)
Nov 24, 2023

[More news >>](#)



Helpdesk

Short- and long-term support with bioinformatics analyses, programming and data management tasks. [Contact the helpdesk.](#)



Services

Analysis and management of life science data within marine, health, genomics, proteomics, structural analysis and more.



NeLS e-infrastructure

NeLS, the Norwegian e-Infrastructure for Life Sciences, for analysis, sharing and storage of high throughput genomics data.



Training

Local and national training in bioinformatics analysis and use of our tools.



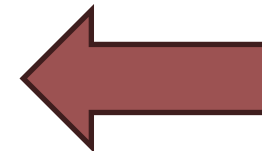
Norges miljø- og biovitenskapelige universitet



The Research Council of Norway

<https://elixir.no/>

Scan me!



Data storage in NeLS and StoreBioInfo (SBI)

Norwegian e-infrastructure for Life Sciences - developed and operated by Elixir Norway

NeLS in general

Available to all Norwegian researchers

Enable collaborative projects

Multiple storage layers

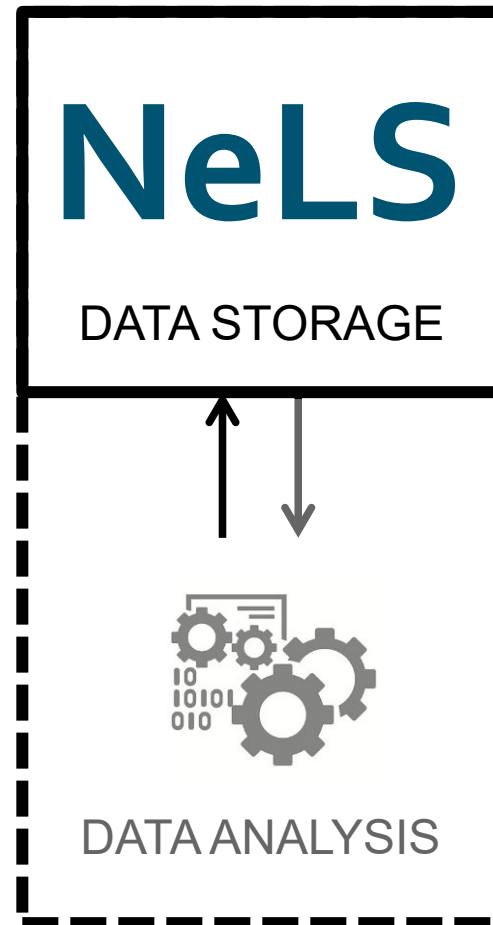
Integrated with compute

Multiple permission roles - shared data

Free of charge*

Support for data deposition

User support: Elixir Norway helpdesk



Type of data

Non-sensitive data

Support multiple data types

SEEK integration for metadata

NeLS

Norwegian e-Infrastructure for Life Sciences



SERVICES

ELIXIR Norway offers services and infrastructure for life science, including software tools and workflows, databases and storage. In addition ELIXIR Norway provides support through its national helpdesk and offers trainings. Moreover, ELIXIR Norway provides services to the international research community.

E-infrastructure

NeLS

Galaxy

TSD@UiO

Other sensitive data platforms

The [Norwegian e-infrastructure for life sciences \(NeLS\)](#),
is a unique national portal for sharing, analysis and
midterm storage of **non-sensitive** molecular life science data.

Free up to 10Tb

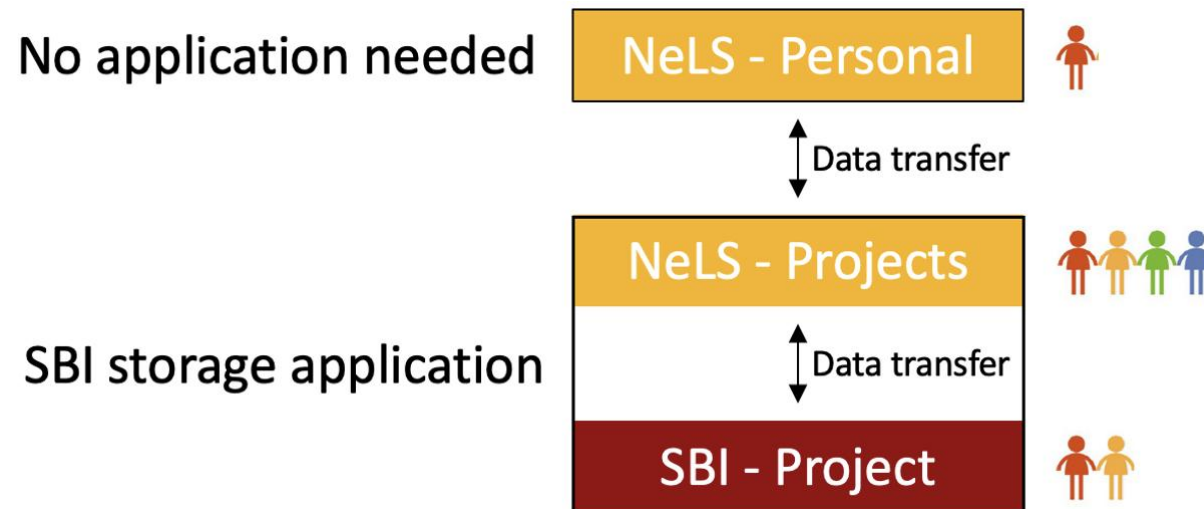


Data storage in NeLS/SBI – How to get access

Storage application needed for projects where data is shared by many users

Access via FEIDE user or NeLS idp can be made for non-FEIDE users

Free storage < 10 TB



Apply for storage: contact@bioinfo.no

Wiki for usage: <https://nels-docs.readthedocs.io/en/latest/>

Login to NeLS using the WebPortal

<https://nels.elixir.no>

NeLS by 

Login

Welcome

This is web portal for Norwegian e-Infrastructure for Life Sciences.

 **Helpdesk**

Short- and long-term support with bioinformatics analyses, programming and data management tasks. [Contact the helpdesk](#)

Read more . . .

 **Services**

Analysis and management of life science data within marine, health, genomics, proteomics, structural analysis and more.

Read more . . .

 **NeLS e-infrastructure**

NeLS, the Norwegian e-Infrastructure for Life Sciences, for analysis, sharing and storage of high throughput genomics data.

Read more . . .

 **Training**

Local and national training in bioinformatics analysis and use of our tools.

Read more . . .

Login

You need a valid user credential in order to access NeLS resources. Select your identity provider:

Login with FEiDE Identity

Norwegian Universities

All Norwegian universities and many other research and education institutions provide a federated login mechanism to allow their users to utilize their own home institution login credentials to access approved online web application services. [Read More . . .](#)

Login with NeLS Identity

Other Norwegian Users and International Collaborators

For users not having access to a Norwegian university account or other academic institution utilizing FEiDE, we provide our own NeLS identity service for both national and international users.

Log in with Feide

NeLS

You need to log in via Feide to access NeLS Portal.

Your affiliation



NMBU Norwegian University of Life Sciences
[Change affiliation](#)



Use work or school account

All users with FEiDE login
credentials can access the NeLS
Portal

<https://nels.elixir.no/>

I don't have FEIDE login credentials (Non-Norwegian collaborators)

- It is also possible to login to NeLS via a NeLS idp. You can apply for a NeLS idp

NeLS

Norwegian e-Infrastructure for Life Sciences

Enter your username and password

A service has requested you to authenticate yourself. Please enter your username and password in the form below.

 Username

 Password

[Login](#) [Forgot Password ?](#)

[Apply for a NeLS Account](#) [Help \(Support\)](#)

Request for NeLS Account

First Name*

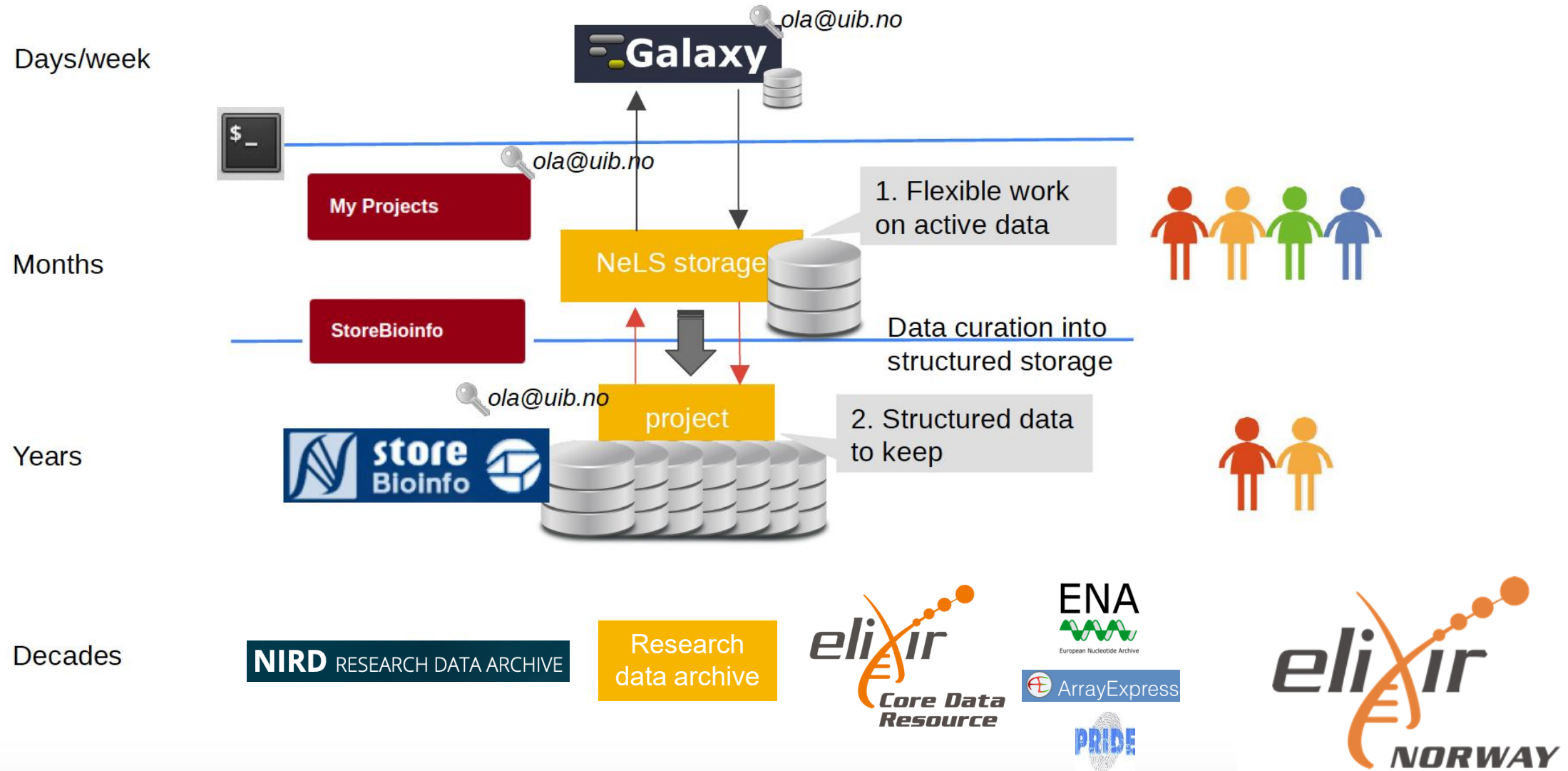
Last Name*

Email* (note: email needs to be home institution email address, not personal email)

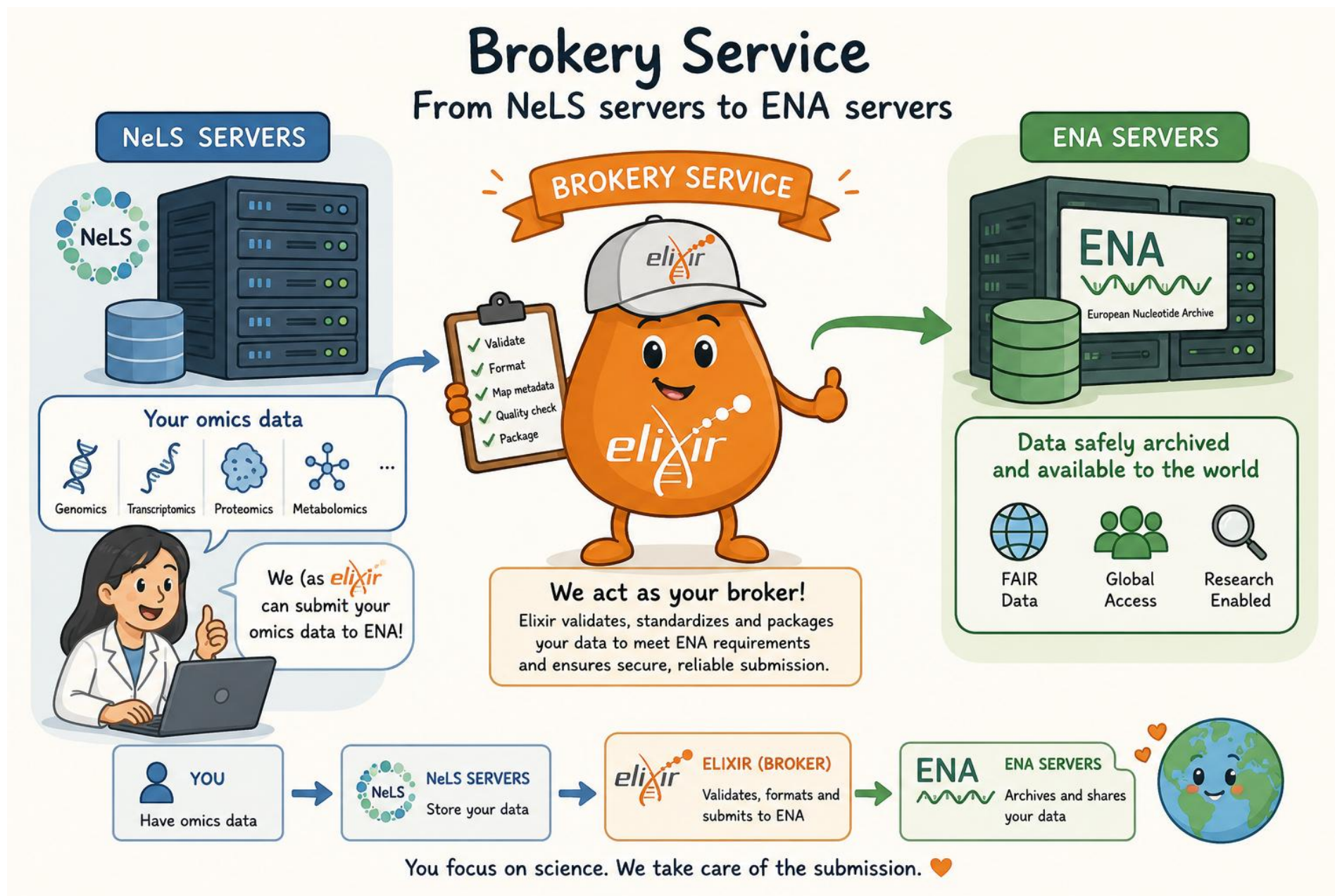
Affiliation*

Usage Description*

NeLS architecture



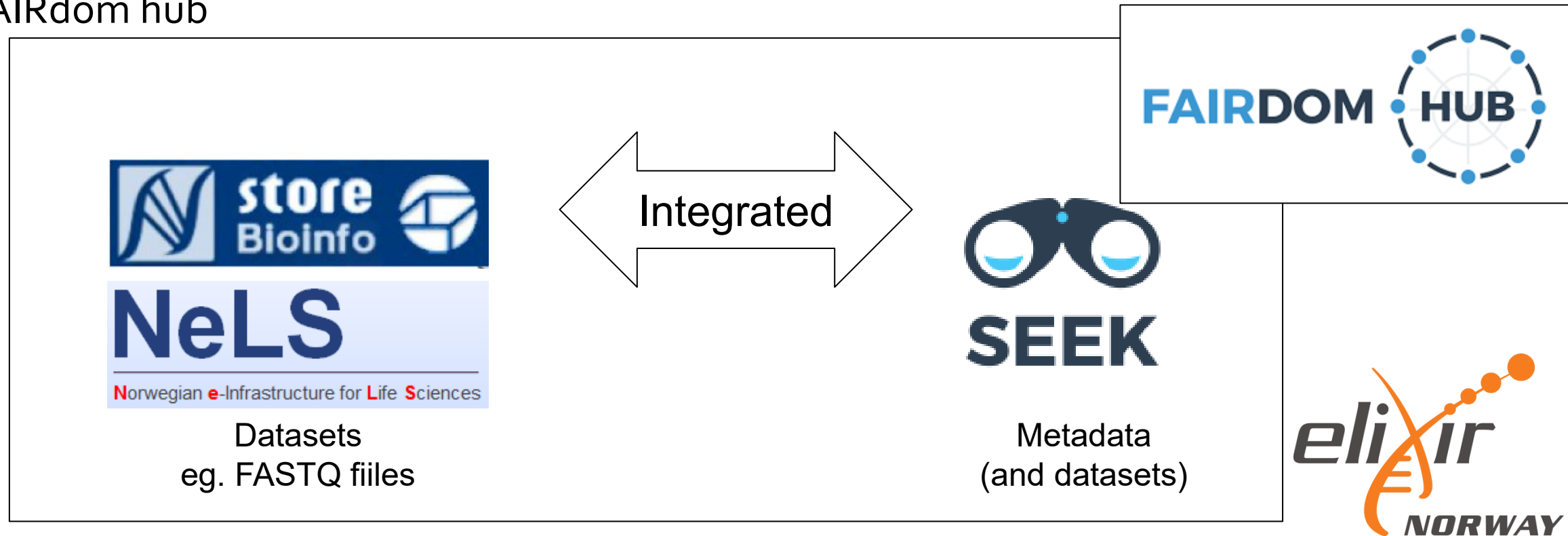
ENA-submission “brokery” service.



SEEK - sharing heterogeneous scientific research datasets, models or simulations, processes and research outcomes

The SEEK platform is a web-based tool for organising and storing data, and for exploring and annotating data

Norwegian users can link datasets stored in NeLS to a SEEK project using FAIRdom hub



Data storage in NIRD

National e-Infrastructure for Research Data - owned and operated by UNINETT Sigma2

NIRD in general

Available to all users with account

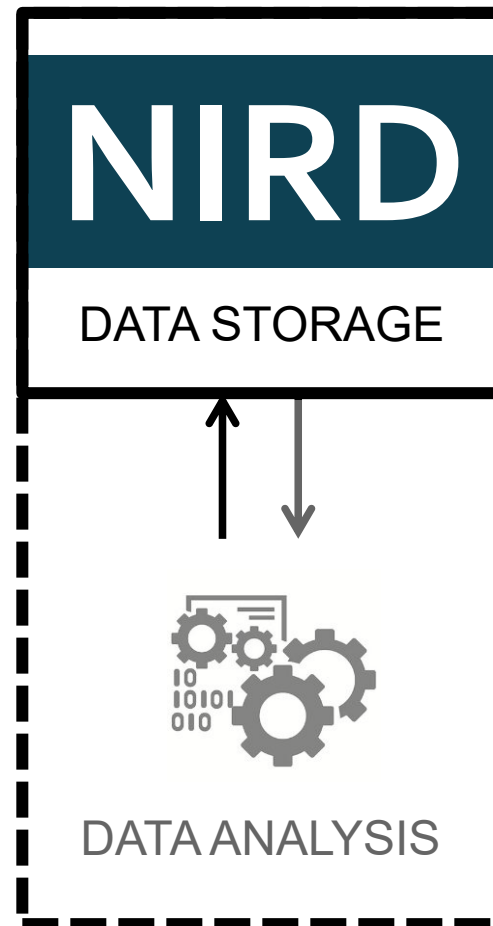
Enable collaborative projects

Integrated with compute

Multiple permission roles - shared data

Free of charge*

User support: Metacenter support

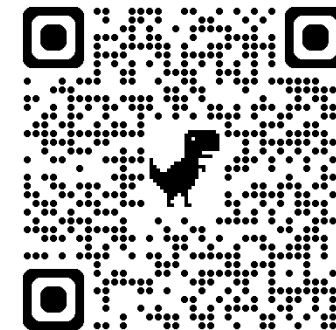


Type of data

Non-sensitive data

Support multiple data types

All type of scientific data*

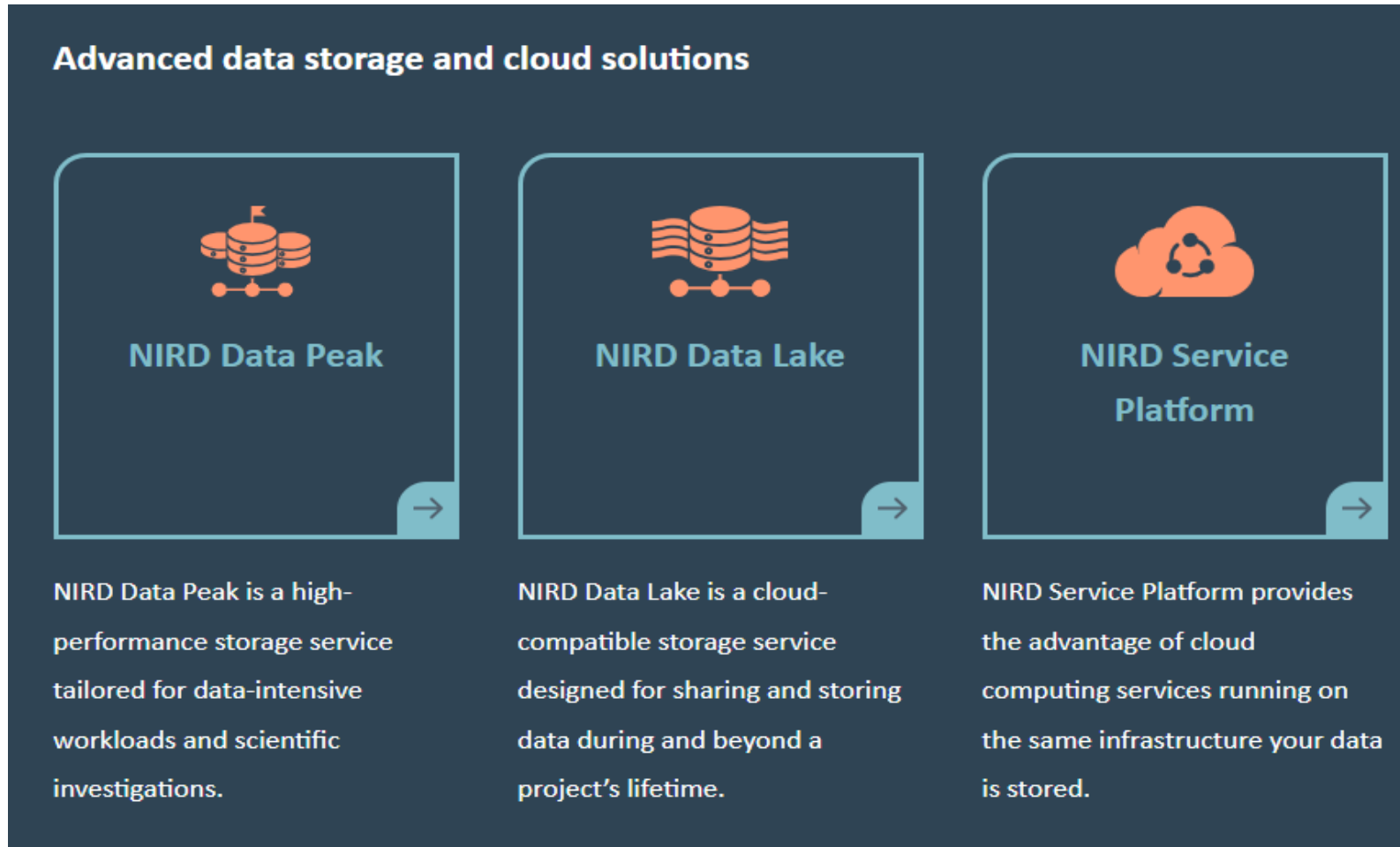


Scan me!



Data storage in NIRD

National e-Infrastructure for Research Data - owned and operated by UNINETT Sigma2



intensive I/O operations

Long term-storage

NIRDTolKit

Data storage in NIRD – How to get access

Apply for a UNINETT SIGMA2 user

Four price categories

The model introduces four different categories of contributors.

- **Category A**
Large projects with funding from the Norwegian Research Council of 15 MNOK or more or storage needs above 10 TiB, paying for operational expenses
- **Category B**
Non-commercial projects needing Dedicated Resources, paying for capital and operational expenses
- **Category C**
Commercial research and industry which will pay the full cost price
- **Category 0**
Non-contributing projects which are smaller projects not in any of the above categories

The process explained

- 1 Submit user application
- 2 Signature by project management
- 3 Review by Sigma2
- 4 Set a password and log in

<https://www.sigma2.no/apply-e-infrastructure-resources>

Scan me!



Access data storage in NIRD

The NIRD storage can be accessed via the command line using SSH

Data import/export via command line tools such as SCP or SFTP or other file transfer tools (e.g. WinSCP)

```
(base) auve@L004538:~$ NIRD
Last login: Wed Nov  6 20:22:33 2024 from 158.36.102.145
Welcome to nird.sigma2.no!

Documentation: https://documentation.sigma2.no/
Support email: support@metacenter.no
Request resources:
                  https://www.sigma2.no/content/apply-e-infrastructure-resources/
~~~~~
Latest news from: https://opslog.sigma2.no

o 2024-11-25: Reduced capacity due to NTNU cooling upgrade week 48
o 2024-11-05: Slowness on Saga
o 2024-10-30: Saga is partially down and very slow.
o 2024-10-28: NIRD datalake full
o 2024-10-21: Issues with login to Fram
~~~~~
Module command is not yet stable. Please report any issue encountered.
~~~~~
(miniforge3) [auve@login0-nird-lmd: ~]$ █
```



NIRD ToolKit

What is the NIRD Service Platform

NIRD Service Platform is a platform to run cloud services, such as web services, domain and community-specific portals, and tools for data visualization, pre/post-processing, data discovery and data sharing.

The service platform offers dedicated resources for data sharing and computing with large datasets. Here you can effortlessly process and share data with your preferred tools, such as

- Jupyter Notebook
- Jupyter Hub
- Spark
- R-Studio

NIRD Toolkit

Filter packages

deep-learning-tools



GPU enabled Jupyter notebook

Read more

Install

desktop-vnc



VNC Ubuntu

Read more

Install

jupyter



Apache Spark Scala R VS Code

Read more

Install

jupyterhub



Jupyter notebook server

Read more

Install

minio



File sharing Cloud storage

Read more

Install

rstudio



R web IDE

Read more

Install

spark



Distributed computing

Read more

Install

<https://apps.sigma2.no/>

NIRD ToolKit: Run Rstudio to analyze data directly on NIRD



NIRD Toolkit

Applications

Arturo Vera Ponce De Leon

Log out

rmemo

Running

> Overview

> Modify

> Resources

Overview

URL(s) rmemo.nirdmemo.sigma2.no
shiny-rmemo.nirdmemo.sigma2.no
Contact email arturo.vera.ponce.de.leon@nmbu.no
Projectspace nirdmemo-ns9864k
Resource usage 2.3 CPU, 8.21 GB RAM

rstudio



RStudio is an integrated development e...
1.3.1

▼ Mounted volumes

Secret

ConfigMap

Temporary storage

Persistent storage

Volume name

oidconfig

Mount path

/conf



Sensitive data storage



Data storage in TSD

National service for sensitive data - developed and operated by UiO

TSD in general

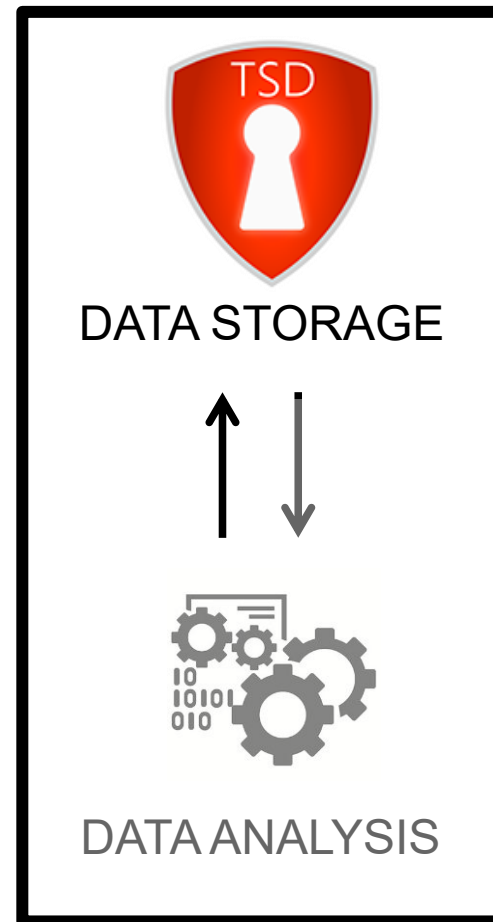
Available to all users with account

Enable collaborative projects

Integrated with compute

Multiple permission roles - shared data

User support: Tech and admin support



Type of data

Sensitive data

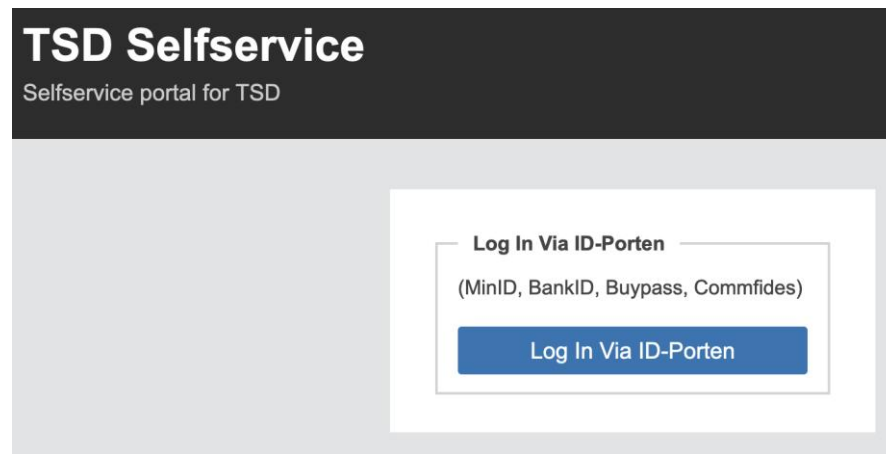
Support multiple data types

Data storage in TSD – How to get access

Apply for a project – need to document ethical approval (e.g. from REC)

Generate a TSD user – require 2-factor authentication

Payment models for storage



Apply for a TSD project through ELIXIR.NO:
ELIXIR Norway has a quota in TSD that, and
users can get subsidised quota through
ELIXIR in TSD

Generate user account: <https://selfservice.tsd.usit.no/>

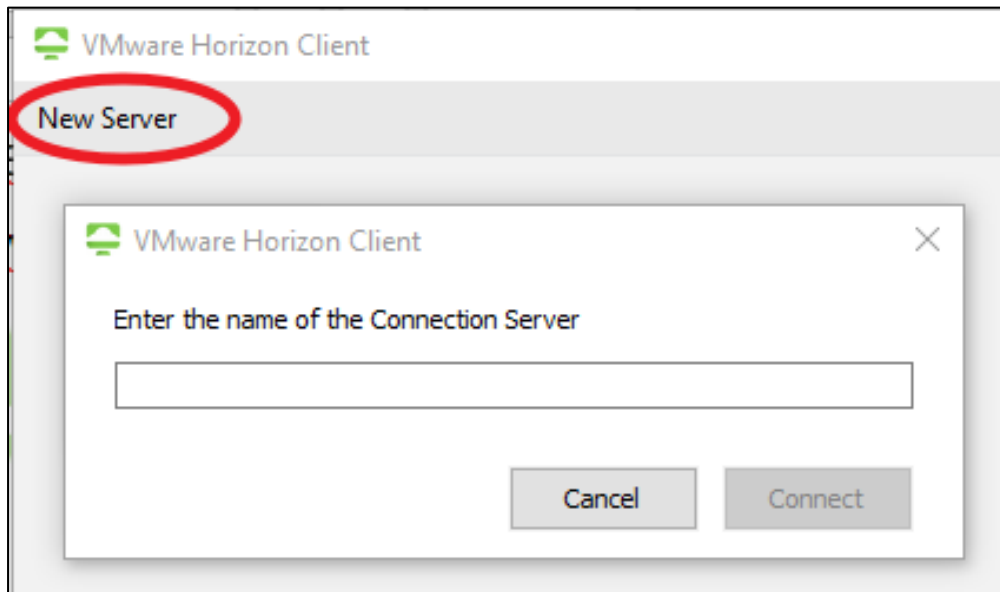
Apply for project: <https://www.uio.no/tjenester/it/forskning/sensitiv/hjelp/start/registrer.html>

Wiki for usage: <https://www.uio.no/tjenester/it/forskning/sensitiv/hjelp/start/index.html>

Access to data storage in TSD

The TSD storage can be accessed via VMware Horizon + 2-factor authentication

Import/export data via the web file upload service



TSD Data Portal

Welcome to the TSD Data Portal

To use this service you should already hold a valid TSD user account.

 Import Files

 Export Files



Where should I store and share my life science data?

	Sensitive data	Direct transfer from data producer	Web portal access*	Command line access	Shared projects	Integration for metadata	Support for data deposition	Free of charge
NeLS/SBI	✗	✓	✓	✓	✓	✓	✓	✗
NIRD	✗	✓	✗	✓	✓	✗	✗	✗
TSD	✓	✓	✗	✓	✓	✗	✗	✗

*Direct access to the data and project management through the web browser

Nothing in life is free...Costs

Category A: Non-commercial projects with allocation from our Resource Allocation Committee (RFK)

This category applies for non-commercial projects with allocation from RFK. Such projects are required to contribute to the operational expenses of the national e-infrastructure services.

Use this category in applications to the Norwegian Research Council and other funders, except the EU.

Prices for category A

Service	Price (valid from 1 October 2025)
HPC CPU	0.04 NOK per CPU-hour
HPC GPU (except LUMI)*	5.6 NOK per GPU hour
NIRD Data Peak	800 NOK per TB/year
NIRD Data Lake	470 NOK per TB/year
Backup	470 NOK per TB/year
Central Data Library	0 NOK** per TB/year

All prices are stated in Norwegian kroner (NOK).

Category B: Non-commercial projects in need of dedicated resources

This category applies for non-commercial projects in need of dedicated resources, paying for capital and operational expenses.

Prices for category B

Service	Price
HPC CPU	0.08 NOK per CPU-hour
HPC GPU*	15.2 NOK per GPU-hour
NIRD Data Peak	1 170 NOK per TB/year
NIRD Data Lake	860 NOK per TB/year
Backup	860 NOK per TB/year
Central Data Library	0 NOK** per TB/year

All prices are stated in Norwegian kroner (NOK).

Category C: Commercial research and industry

This category applies for commercial and research and industry users paying market price.

Prices for category C

Service	Price
HPC CPU	0.13 NOK per CPU-hour
HPC GPU (except LUMI)*	35 NOK per GPU-hour
NIRD Data Peak	1 520 NOK per TB/year
NIRD Data Lake	1 120 NOK per TB/year
Backup	1 120 NOK per TB/year
Central Data Library	1 120 NOK** per TB/year

All prices are stated in Norwegian kroner (NOK).

Example - Data flow/handle using ELIXIR Norway

